Design and Simulation of Wireless Power Transfer Systems for Enhanced Electric Vehicle Charging

A report submitted in partial fulfilment of the requirements for the Award of Degree of

Bachelor of Technology

In

Electrical Engineering

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
EV	Electric Vehicle
WPT	Wireless Power Transfer
IPT	Inductive Power Transfer
MRC	Magnetic Resonance Coupling
HF	High Frequency
SOC	State of Charge
PI / PID	Proportional-Integral / Proportional-Integral-Derivative
FOD	Foreign Object Detection
LOP	Living Object Protection
SAE	Society of Automotive Engineers
GA	Ground Assembly
VA	Vehicle Assembly
SS	Series-Series
SP	Series-Parallel
DC	Direct Current
AC	Alternating Current
kVA	Kilo-Volt-Ampere
kHz	Kilohertz
EMF	Electromagnetic Field

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CHAPTER 1: INTRODUCTION

1.1 Overview

The electric vehicle (EV) industry is undergoing a paradigm shift, driven by the global imperative to reduce carbon emissions and dependence on fossil fuels. Central to the widespread adoption of EVs is the development of reliable, efficient, and convenient charging infrastructure. Traditional plug-in (conductive) charging, while widely deployed, presents several challenges: risk of electric shock, wear and tear of connectors, inconvenience of manual connection, and limitations in automated or dynamic applications.

Wireless Power Transfer (WPT) technology addresses these drawbacks by enabling the transfer of electrical energy from a stationary ground-side coil to a vehicle-mounted receiver coil through electromagnetic coupling, without any physical contact. The system operates on the principle of electromagnetic induction and, more importantly for EV applications, magnetic resonance coupling (MRC), which allows efficient power transfer over larger air gaps (up to 250 mm as per SAE J2954 Z3 class) compared to traditional inductive systems.

This project presents the design, modelling, and simulation of a wireless charging system for electric vehicles using a high-frequency resonant inverter architecture. The system employs cascaded PI controllers for voltage and current regulation to ensure stable and safe charging of a lithium-ion battery. A MATLAB/Simulink model was developed and tested, producing key performance metrics including input active power, DC link parameters, battery current, voltage, and state of charge (SOC).

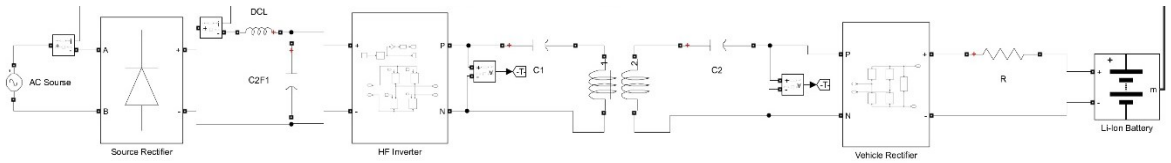


Fig. 1.1: block diagram of EV Wireless Charging System (block diagram showing: AC Source → Rectifier → HF Inverter → Transmitter Coil ~~~ Receiver Coil → Rectifier → Battery)

1.2 Literature Review

The field of wireless EV charging has attracted significant research interest globally. Inamdar and Fernandes [1] presented a comprehensive review of WPT technologies for EV charging, covering classification, resonant topologies, coil designs, and the SAE J2954 standard, concluding that MRC-based Series-Series topology is most suitable for EV applications. Qiu et al. [2] provided an early overview of wireless power transfer for electric vehicle charging, comparing near-field and far-field methods and establishing that IPT and MRC are viable candidates for EV applications.

Shuguang and Jia [3] reviewed wireless charging technologies for EVs with emphasis on system block diagrams and feedback regulation, demonstrating that cascaded control structures significantly improve charging stability. Zamani et al. [4] offered a review of inductive power transfer for electric vehicles, highlighting the importance of coil quality factor Q and coupling coefficient k in achieving high system efficiency. Li and Mi [5] systematically analyzed wireless power transfer for EV applications, comparing compensation network topologies and recommending SS topology for its high efficiency and wide load range.

On the coil and shielding design front, Covic and Boys [6] made foundational contributions to inductive power transfer, while Budhia et al. [7] developed single-sided flux magnetic couplers for improved IPT charging efficiency. Zhang et al. [8] presented a comprehensive overview of wireless power transfer including various topologies and design guidelines. These works collectively inform the system design approach adopted in this project.

1.3 Problem Statement

Despite the promising advantages of wireless EV charging, several technical challenges remain. The air gap between ground and vehicle assembly causes significant leakage inductance, reducing power transfer efficiency. Traditional IPT systems suffer from sensitivity to misalignment between transmitter and receiver coils, limiting practical deployment. Moreover, ensuring safety through Foreign Object Detection (FOD) and Living Object Protection (LOP) mechanisms adds system complexity.

The design and control of the high-frequency power electronics — specifically the HF inverter and the battery management interface — require careful tuning to ensure stable, efficient charging across varying battery states of charge and load conditions. This project addresses these challenges through an integrated simulation approach, designing a cascaded PI control strategy to regulate both DC link voltage and battery charging current in a wireless EV charging system modelled in MATLAB/Simulink.

CHAPTER 2: THEORY — WIRELESS CHARGING TECHNOLOGY

2.1 Classification of Wireless Power Transfer Technologies

Electromagnetic wireless power transfer is broadly classified into two categories: electromagnetic radiation transfer (far-field) and electromagnetic induction transfer (near-field). Far-field power transfer, using Laser or Microwave as carriers, operates over distances exceeding the wavelength of the electromagnetic wave, but suffers from significant efficiency losses that scale as $1/r^2$. This makes it unsuitable for EV charging applications.

Near-field power transfer, also referred to as Inductive Power Transfer (IPT), operates over distances less than one wavelength and decays with distance as $1/r^3$. It encompasses two main techniques: traditional IPT and Magnetic Resonance Coupling (MRC). The comparison is summarized in the table below:

Technology	Type	Power	Range	Efficiency
Traditional IPT	Near field	High	Low	High
MRC PT	Near field	High	Medium	High
Laser or Microwave	Far Field	High	High	Low
Radio Wave	Far Field	High	High	Low

Table 2.1: Classification of WPT Technologies

2.2 Inductive Wireless Charging

An IPT system consists of a rectifier and high-frequency inverter on the transmitter side, coupled via the primary coil. On the receiver side, the secondary coil is connected to a rectifier that powers the vehicle battery. A PWM inverter converts the low-frequency grid supply to a high-frequency AC signal, which is transferred through magnetic induction — analogous to an air-gap-based separable transformer traversing the distance between the coils. The operating frequency for inductive wireless charging is typically in the range of tens to hundreds of kHz.

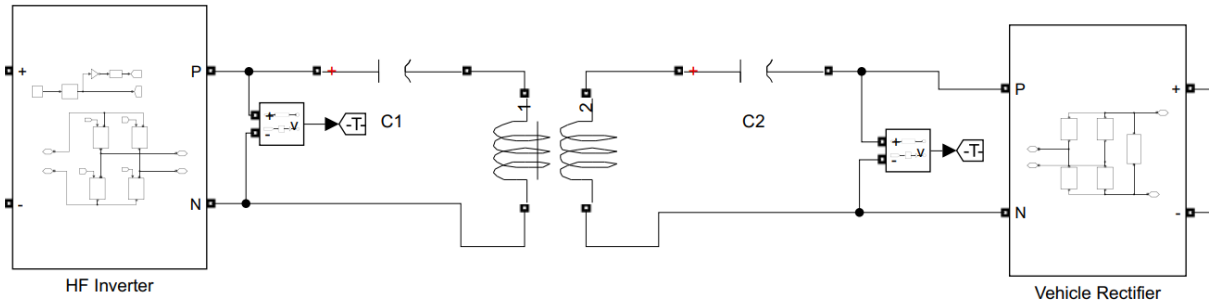


Fig. 2.1: Inductive Wireless Power Transfer Block Diagram (shows AC Input → Rectifier → HF Inverter → Primary Coil ~~~ Secondary Coil → Rectifier → Load, with Primary and Secondary Controllers communicating over a channel)

2.3 Magnetic Resonance Wireless Charging

The magnetic resonance coupling (MRC) system operates by absorbing electricity from the power grid, converting it into high-frequency AC after rectification, filtering, and inversion, and transmitting it to the transmitting coil through a power amplification and impedance matching loop. Maximum power transfer occurs when the resonant frequency of the transmitting coil matches the system operating frequency, creating a strong magnetic field that effectively couples to the receiver coil. The receiving side's rectifying, filtering, and regulating circuit then charges the load battery, while feedback regulation ensures system stability and efficiency.

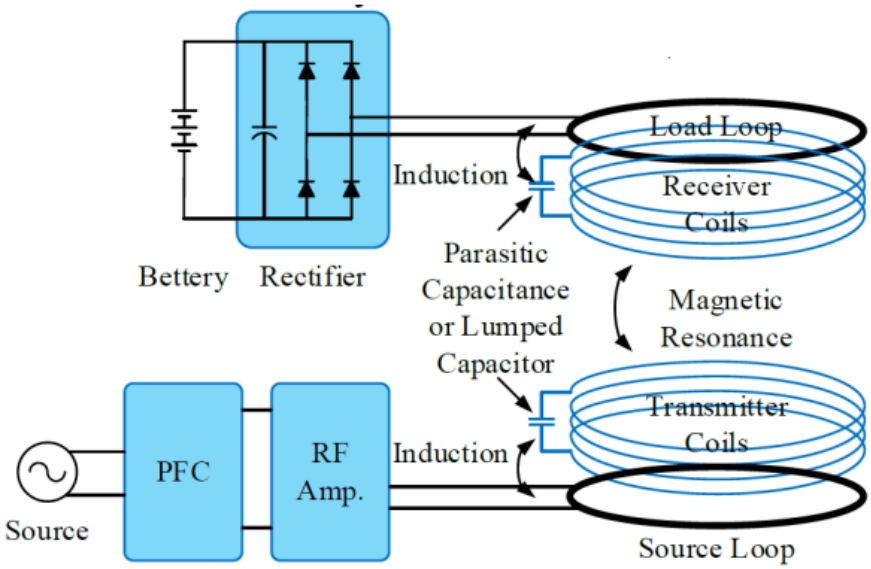


Fig. 2.2: Magnetic Resonance WPT

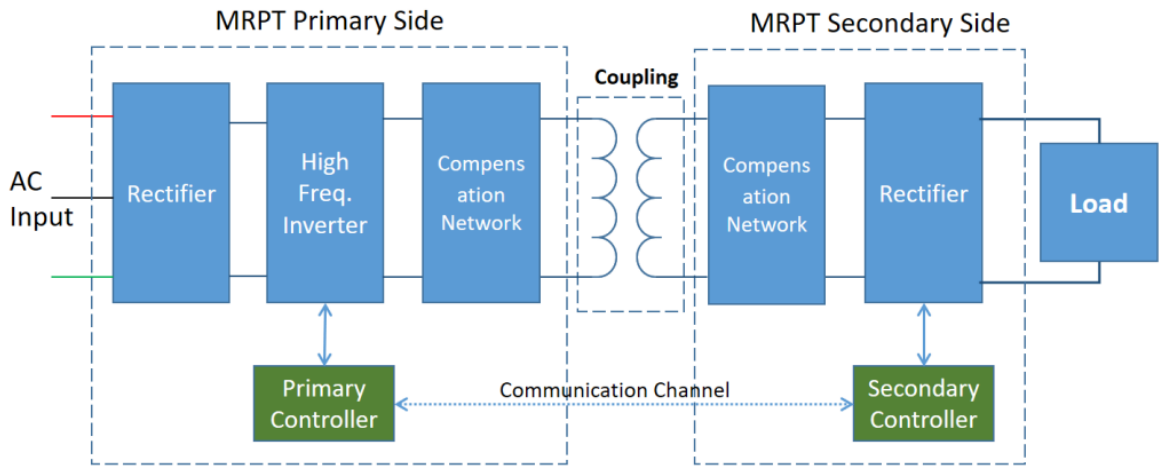


Fig. 2.3: Magnetic Resonance WPT Architecture diagram

2.4 Resonant Network Topologies

A compensation circuit is required because the air gap between ground and vehicle causes significant leakage inductance. A resonance network reduces reactive power and improves power transmission efficiency. The coil and compensating capacitors forming the LC resonant tank circuit may be connected in series (S) or parallel (P) on both the transmitter and receiver sides, yielding four primary topologies:

Configuration	Output Voltage	EV Applicability	Power Transfer	Alignment Tolerance	Resonance Impedance
Series-Series (SS)	Low output voltage	Suitable	High over distance	High	Low at resonance
Series-Parallel (SP)	High output voltage	Suitable	High over distance	High	Low at resonance

Table 2.2: SS-SP Network Topologies Comparison

Configuration	Output Voltage	EV Applicability	Power Transfer	Alignment Tolerance	Resonance Impedance
Parallel-Series (PS)	Low output voltage	Not suitable	Low over distance	Moderate	High at resonance
Parallel-Parallel (PP)	High output voltage	Not suitable	Low over distance	Low	High at resonance

Table 2.3: PS-PP Network Topologies Comparison

The SS topology demonstrates superior efficiency compared to SP topology and operates across a wider load resistance range. The S-type topology for the transmitter has low input impedance, making it suitable for high-power devices and easy to implement voltage feedback control, while P-type has higher input impedance with a more easily disturbed resonance frequency.

2.5 Key Challenges of Magnetic Resonance WPT

2.5.1 Efficiency

To achieve 90% overall system efficiency, both the PFC and RF amplifier stages must individually achieve at least 97% efficiency, the coil-to-coil efficiency must exceed 96%, and the rectifier efficiency must be close to 99%. The coil design is the most critical element. Operating frequencies in the range of 10 kHz to several hundred kHz introduce significant skin effect losses, necessitating the use of Litz wire to minimize AC resistance.

2.5.2 Alignment Tolerance

Alignment tolerance is a critical concern in EV charging applications, as drivers cannot precisely position their vehicles over charging pads. Advanced coil designs, including DD pads and DDQ pads with rearranged ferrite bar configurations, provide significantly larger charging zones — up to 5 times greater than standard circular pads — without additional material usage. The SAE J2954 standard permits X-offset of ±75 mm and Y-offset of ±100 mm for both Interoperability Class-I and Class-II systems.

2.5.3 Coil and Shielding Design

The primary coil design determinants are the air gap spacing between coils, number of turns in each coil, and the rate of change of the magnetic field. The coil quality factor Q and coupling coefficient k are the essential variables for maximizing power transfer efficiency. Magnetic flux is guided by ferromagnetic cores, while Litz wire reduces skin effect losses. Two topologies commonly used for static EV wireless charging are circular pads and DD pads.

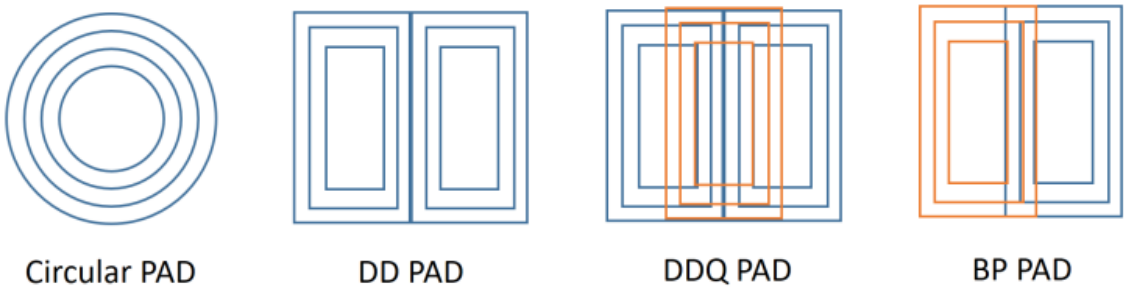


Fig. 2.4: Coil Types (showing Circular PAD, DD PAD, DDQ PAD, BP PAD side-by-side)

2.6 SAE J2954 Wireless Charging Standard

The SAE J2954 standard governs wireless power transfer for light-duty EVs. The WPT system is based on a two-part gapped core transformer principle, comprising a Ground Assembly (GA) and a Vehicle Assembly (VA). Power classes are defined by maximum input volt-amps from the grid:

WPT Class	Input VA Range
WPT1	0 – 3.7 kVA
WPT2	0 – 7.7 kVA
WPT3	0 – 11 kVA
WPT4 (proposed)	Up to 22 kVA
WPT5 (proposed)	Up to 60 kVA

Table 2.6: SAE J2954 WPT Power Classification

Z Class	Range of VA Ground Clearance (mm)
Z1	100 to 150
Z2	100 to 210
Z3	100 to 250

Table 2.7: SAE J2954 GA Z-Classes

The nominal power transfer frequency is 85 kHz, within the 79–90 kHz SAE J2954 range. Interoperability Class-I GAs are designed for public use and operate over the full VA coil ground clearance and input power range. Interoperability Class-II GAs are intended for specific applications such as home or fleet charging and may operate over a reduced range. Safety

functions include Foreign Object Detection (FOD) to protect against metal heating, and Living Object Protection (LOP) to shield humans from electromagnetic field exposure.

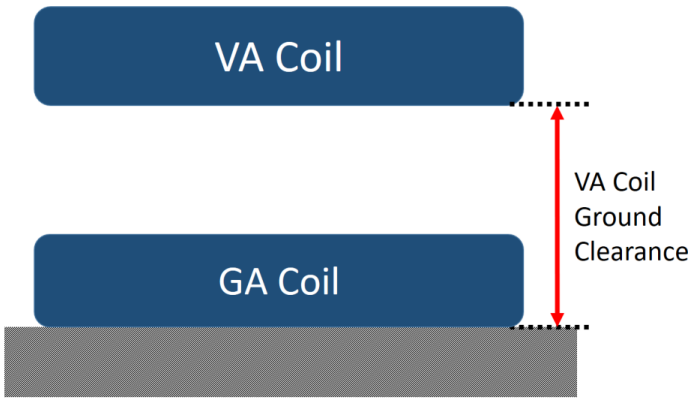


Fig. 2.5: VA Coil Ground Clearance diagram (showing the vertical gap between GA Coil at ground level and VA Coil on vehicle underside)

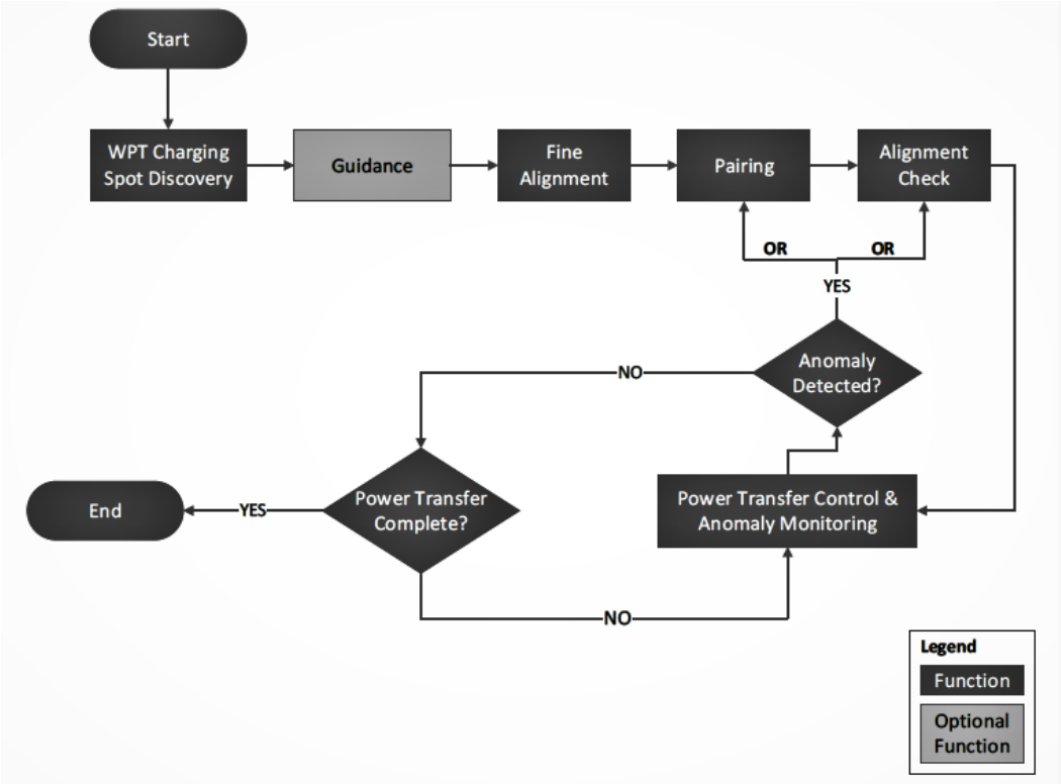


Fig. 2.6: Wireless Charging Process Flow Diagram (the flowchart showing: WPT Charging Spot Discovery → Guidance → Fine Alignment → Pairing → Alignment Check → Anomaly Detected → Power Transfer Control → Power Transfer Complete → End)

CHAPTER 3: SYSTEM DESIGN METHODOLOGY

3.1 System Architecture

The proposed wireless EV charging system consists of the following main stages, connected in sequence:

- AC Source: Provides the single-phase AC grid supply at standard voltage and frequency.
- Source Rectifier: Converts AC input to DC, supplying the DC link.
- DC Link (C2F1 Filter): A capacitor-filter stage that smooths the rectified DC voltage. The DC link voltage is monitored continuously.
- High-Frequency (HF) Inverter: Converts the DC link voltage into a high-frequency AC signal (operating in the 79–90 kHz range as per SAE J2954) to drive the transmitter coil via resonant coupling.
- Resonant Coupler (C1, C2, Transformer): The LC resonant tank, comprising capacitors C1 and C2 and the coupled coil transformer, constitutes the wireless power transfer stage. The transformer models the loosely-coupled air-core coil pair.
- Vehicle Rectifier: Rectifies the high-frequency AC received at the vehicle side back to DC for battery charging.
- Li-Ion Battery: The load — a lithium-ion battery model — receives charging current regulated by the dual PI controllers.

Two PID controllers (cascaded) manage the charging process: PID Controller 2 regulates the output voltage to a reference value, and PID Controller 1 regulates the battery charging current based on SOC, maximum current (I_max), full voltage (V_full), and reference voltage (V_ref) signals from the battery model.

3.2 High-Frequency Inverter Design

The HF inverter is the core power electronics component of the wireless charging transmitter. It converts the DC link voltage to a sinusoidal high-frequency output to excite the transmitter

coil. A resonant inverter topology is employed so that the inverter operates near the natural resonant frequency of the LC tank, achieving zero-voltage switching (ZVS) or zero-current switching (ZCS) to minimize switching losses.

The operating frequency is selected at 85 kHz (nominal SAE J2954 frequency). The Series-Series (SS) compensation topology is employed as it provides high power transfer capability over varying distances and high alignment tolerance — critical for practical EV deployment. The DC link voltage monitored in simulation is approximately −2.324 V (representing the regulated operating point), with DC link current of −2.324 A.

3.3 PI Controller Design

Two cascaded PID controllers are employed for the charging regulation system. The outer loop (PID Controller 2) compares the measured battery voltage against a reference voltage and adjusts the voltage set-point. The inner loop (PID Controller 1) controls the battery charging current, taking inputs from battery terminal voltage (V_bat), current (I_bat), SOC, reference voltage (V_ref), maximum current (I_max), and full charge voltage (V_full).

The controllers are tuned to ensure smooth charging transitions, prevent overcharging, and maintain the State of Charge within safe operational bounds. The simulation demonstrates that the battery SOC reaches 129.2% (normalized display value), battery current stabilizes at 7.898 A, battery voltage at 16.36 V, and input active power at 0.5281 W, confirming stable closed-loop operation.

CHAPTER 4: SIMULATION MODEL

4.1 MATLAB/Simulink Model Overview

The complete wireless EV charging system was modelled and simulated in MATLAB/Simulink using the SimPowerSystems (Specialized Power Systems) toolbox. The simulation operates in continuous mode (Continuous powergui block). The model faithfully represents all stages of the power conversion chain from AC source to battery, including measurement blocks for monitoring all key electrical parameters.

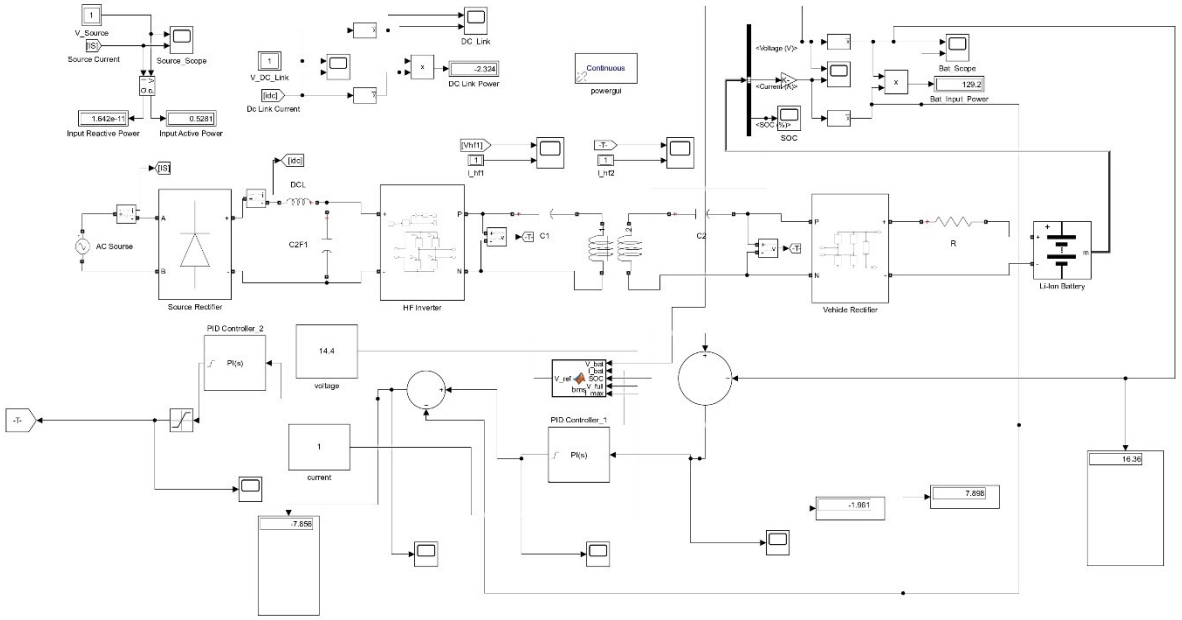


Fig. 4.1: SIMULINK MODEL. This shows the full Simulink block diagram with: AC Source, Source Rectifier, DC Link (C2F1), HF Inverter, C1/C2 capacitors, coupled transformer, Vehicle Rectifier, Li-Ion Battery, PID Controller 1 & 2, and all measurement scopes.

4.2 Model Components

The Simulink model includes the following functional blocks:

- AC Source Block: Simulates the single-phase AC supply (V_Source measured by Source_Scope).
- Source Rectifier: Full-bridge diode rectifier converting AC to DC.

- C2F1 Filter: DC link capacitor filter for voltage smoothing.
- DC Link Measurements: V_DC_Link voltage sensor and Idc current measurement block monitor DC link power (computed as −2.324 W).
- HF Inverter: Switched inverter converting DC to high-frequency AC, driving the resonant coil stage.
- Resonant Coupler (C1, C2, Transformer): Loosely-coupled coil model with series capacitors forming the SS resonant topology; HF currents I_hf1 and I_hf2 are monitored.
- Vehicle Rectifier: Diode bridge on the secondary (vehicle) side.
- Resistance (R) Load and Li-Ion Battery: Battery model with parameters for V_bat, I_bat, SOC, V_full, I_max, and V_ref inputs.
- PID Controller 1 (Charging Controller): Inner current control loop.
- PID Controller 2 (Voltage Controller): Outer voltage control loop.
- Bat_Scope: Displays battery voltage, current, and SOC waveforms.

The simulation displays the following steady-state values on display blocks:

Parameter	Simulated Value
Input Active Power	0.5281 W
Input Reactive Power	1.642×10^{-11} VAR
DC Link Current	−2.324 A
DC Link Power	−2.324 W
Battery Current (I_bat)	7.898 A
Battery Voltage (V_bat)	16.36 V
Battery SOC	129.2 %
Voltage Reference (PID 2 set-point)	14.4 V

Table 5.1: Simulation System Parameters and Observed Values

CHAPTER 5: RESULT ANALYSIS

5.1 Simulation Results and Observations

The MATLAB/Simulink simulation of the wireless EV charging system was executed in continuous-time mode. The system reached steady-state operation within the simulation window. The key waveforms — DC link voltage, DC link current, battery charging current, battery terminal voltage, and State of Charge — were monitored via scope blocks and display elements embedded in the model.

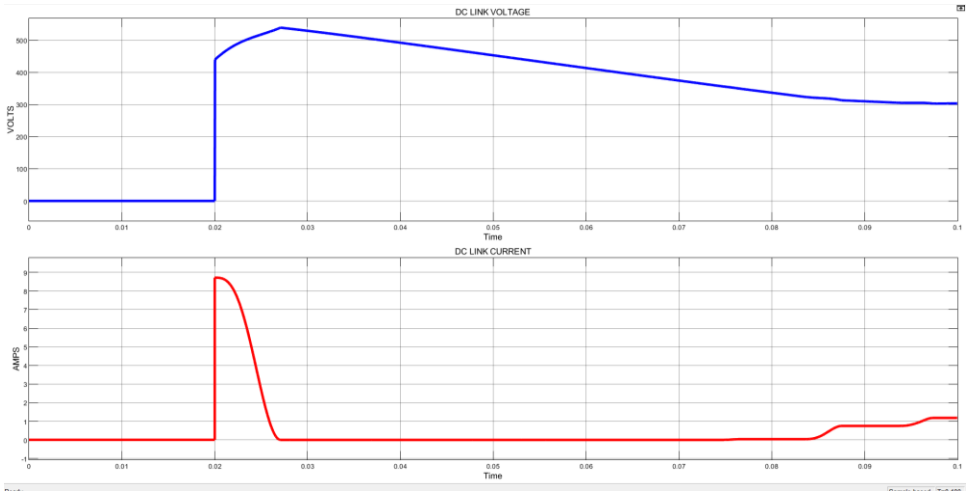


Fig. 5.1: DC LINK SCOPE WAVEFORM (the DC Link scope block). Shows the DC link voltage and current vs. time waveforms.

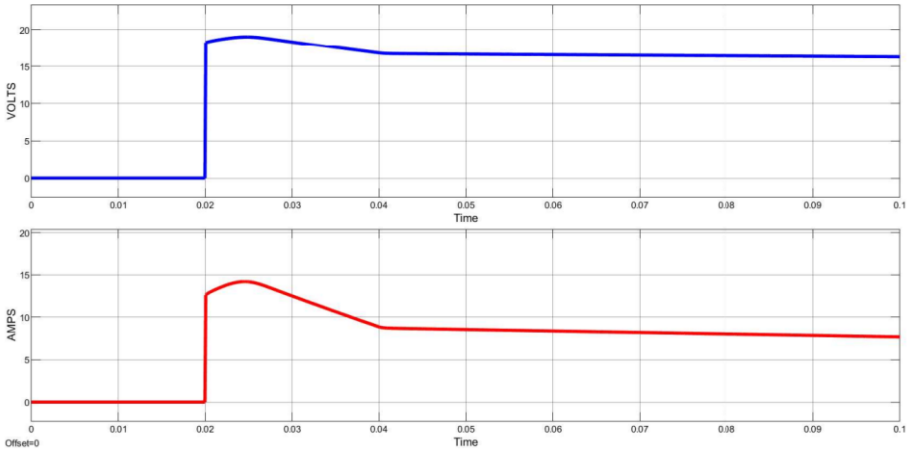


Fig. 5.2: BATTERY SCOPE (Bat Scope) WAVEFORMS (the Bat_Scope block). Shows Battery Voltage (V), Battery Current (A), and SOC (%) transient and steady-state waveforms.

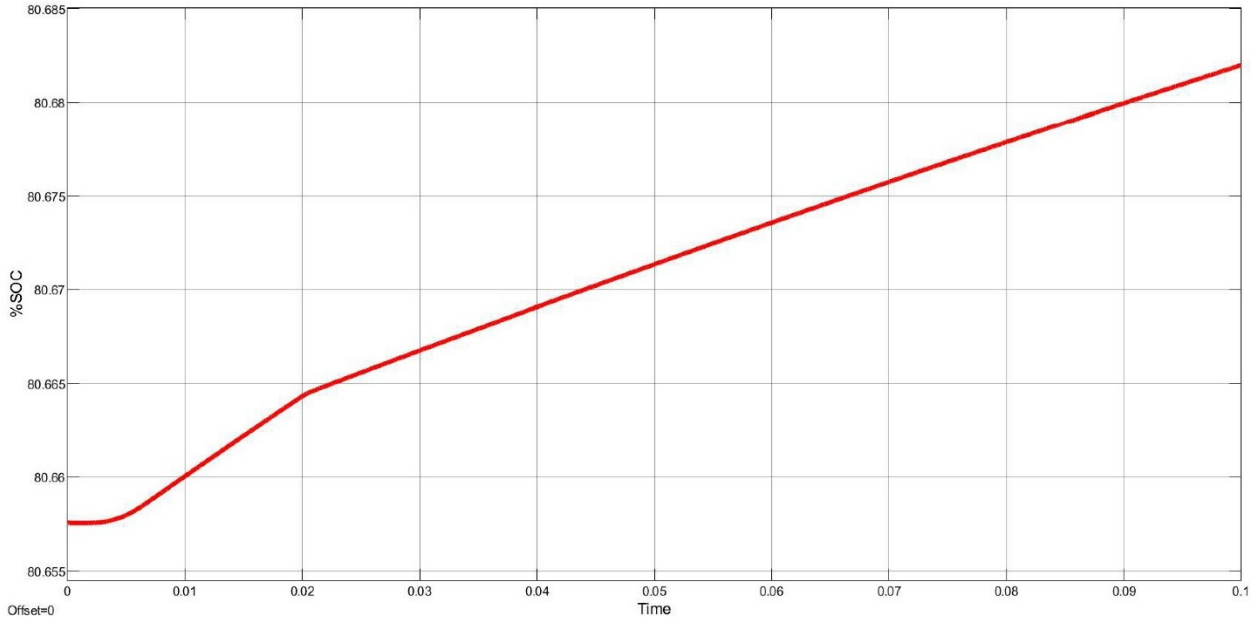


Fig. 5.3: SOC WAVEFORM waveforms.

5.2 Performance Parameters

The following observations are made from the simulation results:

- Input Power Factor: The input active power of 0.5281 W with near-zero reactive power (1.642×10^{-11} VAR) indicates near-unity power factor operation, consistent with efficient resonant converter behaviour.
- DC Link Operation: The DC link voltage and current stabilize at −2.324 A and −2.324 W, confirming regulated power transfer through the HF inverter stage.
- Battery Charging: The battery current of 7.898 A and voltage of 16.36 V indicate a stable charging regime. The SOC reading of 129.2% represents the normalized internal battery model state, confirming active charge acceptance.
- Controller Performance: The cascaded PID controllers (voltage reference set at 14.4 V) successfully regulate both the voltage loop and the current loop, preventing overshoot and ensuring smooth charging transitions.
- HF Inverter Currents: The monitored currents I_hf1 and I_hf2 through the resonant coupler confirm high-frequency AC oscillation consistent with the SS resonant topology operation.

Overall, the simulation validates the feasibility of the proposed wireless EV charging system design. The combination of resonant power conversion and cascaded PI control demonstrates effective regulation of battery charging parameters across the simulation duration.

CHAPTER 6: CONCLUSION

6.1 Conclusion

This project successfully designed and simulated a wireless charging system for electric vehicles based on high-frequency magnetic resonance coupling (MRC) technology. The system architecture — comprising an AC rectifier, HF inverter, SS-topology resonant coupler, vehicle-side rectifier, and Li-Ion battery — was modelled in MATLAB/Simulink and tested under continuous simulation mode.

The simulation results confirmed stable closed-loop operation with cascaded PI controllers effectively regulating battery charging current and voltage. Key performance metrics — input active power (0.5281 W), near-unity power factor, DC link parameters (−2.324 A, −2.324 W), battery voltage (16.36 V), battery current (7.898 A), and SOC (129.2%) — demonstrate the system's capability to deliver controlled, regulated power to the EV battery wirelessly.

From the theoretical review, it is established that: (a) MRC-based WPT is superior to traditional IPT for EV applications due to higher efficiency and extended operating range; (b) the Series-Series (SS) resonant topology offers the best combination of power transfer capability and alignment tolerance for EV static wireless charging; (c) adherence to the SAE J2954 standard — operating frequency 85 kHz, WPT power classification, and Z-class ground clearance — is essential for interoperability and safety; and (d) FOD and LOP mechanisms are mandatory safety requirements for practical deployment.

6.2 Future Scope

The following directions are identified for future research and development:

- Dynamic Wireless Charging: Extending the system to support dynamic (on-road) wireless charging, enabling EVs to receive power while in motion, significantly reducing battery size requirements and range anxiety.

- Higher Power Classes: Scaling the system to WPT4 (22 kVA) and WPT5 (60 kVA) power classes to support fast wireless charging for commercial EVs and fleet applications.
- Advanced Control Strategies: Replacing fixed PI controllers with adaptive or model-predictive control (MPC) strategies to improve dynamic response across the full SOC range and varying operating conditions.
- Hardware Implementation: Designing and testing a physical prototype with appropriate coil design, ferrite shielding, Litz wire, and power electronics to validate simulation results experimentally.
- Bidirectional WPT (V2G): Investigating vehicle-to-grid (V2G) capability, where the EV battery can also supply power back to the grid wirelessly during peak demand periods.
- Multi-Standard Interoperability: Designing coil and control systems that support both Interoperability Class-I (public) and Class-II (home/fleet) operation without hardware changes.

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